2022

T40 & T20P
Repair Training

Contents

THE FUTURE OF POSSIBLE

01 Product Introduction

01 Product Introduction

Product Name: T40/T20P Agras Drone

- Equipped with a coaxial twin-rotor design, the T40 boasts a spraying capacity of 40 kg.
- The lightweight and foldable T20P agricultural drone supports a spreading capacity of 25 kg and a spraying capacity of 20 kg.
- Equipped with dual atomization spraying system, DJI Terra, active phased array radar and binocular vision system, both new drones support aerial surveying and mapping functions.

01 Product Introduction — Product Comparison

T40

- 40L Detachable Spray Tank
- Coaxial Twin-Rotor
- Active Phased Array Radar
- Dual Vision Sensors
- Weight Sensor (real-time remaining liquid amount)
- Single Point Liquid Level Gauge (Makes alert when tank is empty)
- Ultra HD Camera and Adjustable Gimbal Angle
- IPX6K (resist water jets of extremely high pressure)
- Dual Centrifugal Atomization Nozzle
- Magnetic Drive Impeller Pump
- Agricultural Application Scenarios such as Precision Spraying, Spreading, Aerial Surveying, and Mapping

T30

- 30L Fixed Spray Tank
- Spherical Omnidirectional Radar System
- Continuous Liquid Level Gauge (with real-time pesticide load detection and intelligent supply-point prediction)
- Dual FPV Cameras for Real-Time Monitoring
- IP67 Rating
- High-Pressure Nozzle
- Plunger Pump

01 Product Introduction — Product Comparison

T20P

- 20L Detachable Spray Tank
- Active Phased Array Radar
- Dual Vision Sensors
- Weight Sensor (real-time remaining liquid amount)
- Single Point Liquid Level Gauge (Makes alert when tank is empty)
- Ultra HD Camera and Adjustable Gimbal Angle
- IPX6K (resist water jets of extremely high pressure)
- Dual Centrifugal Atomization Nozzle
- Magnetic Drive Impeller Pump
- Agricultural Application Scenarios such as Precision Spraying, Spreading, Aerial Surveying, and Mapping

T10

- 10L Detachable Spray Tank
- Spherical Omnidirectional Radar System
- Single Point Liquid Level Gauge (Makes alert when tank is empty)
- Dual FPV Cameras for Real-Time Monitoring
- IP67 Rating
- High-Pressure Nozzle
- Diaphragm Pump

01 Product Introduction — Specifications

		T40	T30
Features	Hourly Work Efficiency	57 acres	40 acres
Drone Parameters	Total Weight (without batteries)	38 kg	26.3 kg
	Maximum Take-off Weight	90 kg (near sea level)	76.5 kg (near sea level)
	Maximum Rotor Distance	2160 mm	2145 mm
	Dimensions	2,800 mm × 3,150 mm × 820 mm (with arms and blades unfolded) 1,580 mm × 1,930 mm × 820 mm (with arms unfolded and blades folded) 1,150 mm × 760 mm × 870 mm (with arms folded)	2,858 mm × 2,685 mm × 790 mm (with arms and blades unfolded) 2,030 mm × 1,866 mm × 790 mm (with arms unfolded and blades folded) 1,170 mm × 670 mm × 857 mm (with arms folded)
	Hovering Precision (with good GNSS signal)	With D-RTK enabled: ±10 cm (horizontal) and ±10 cm (vertical) With D-RTK disabled: ±0.6m (horizontal) and ±0.3m (vertical) (with the radar function enabled: ±0.1m)	With D-RTK enabled: ±10 cm (horizontal) and ±10 cm (vertical) With D-RTK disabled: ±0.6m (horizontal) and ±0.3m (vertical) (with the radar function enabled: ±0.1m)
	Hovering Endurance	18.5 min (@30,000 mAh & takeoff weight of 50 kg) 7.5 min (@30,000 mAh & takeoff weight of 90 kg)	20.5 min (@29,000 mAh & takeoff weight of 36.5 kg) 7.8 min (@29,000 mAh & takeoff weight of 66.5 kg)
Power System	Stator Size	100×33 mm	100×18 mm
	Maximum Power	4000 W/rotor	3600 W/rotor
	Propeller Diameter	54 inch	38 inch
	Rotor Quantity	8	6
Operation Tank	Operation Tank Volume	40L at full load	30L at full load
	Operation Load	40 kg at full load	30 kg at full load
Nozzles	Nozzle Model	LX8060SZ: 12 L/min	SX11001VS (standard): 7.2 L/min SX110015VS (optional) 8 L/min Fruit tree drones: TX-VK04 (optional): 3.6 L/min
	Nozzle Quantity:	2	16
	Atomized Particle Size	50 - 300 μm	SX11001VS: 130 - 250 μm SX110015VS: 170 - 265 μm TX-VK4: 110 - 135 μm
	Maximum Effective Spray Width	11 m (with a distance of 2.5 meters to crops)	4-9 m (with 12 nozzles and a distance of 1.5 to 3 meters to crops)
Water Pump	Model	Magnetic Drive Impeller Pump	Plunger Pump
	Maximum Flow	6 L/min x 2	4L/min ×1

01 Product Introduction — Specifications

		T20P	T10
Features	Hourly Work Efficiency	30 acres	15 acres
Drone Parameters	Total Weight (without batteries)	25.2 kg	13 kg
	Maximum Take-off Weight	51 kg (near sea level)	26.8 kg (near sea level)
	Maximum Rotor Distance	2192 mm	1480 mm
	Dimensions	2,800 mm × 3,120 mm × 640 mm (with arms and blades unfolded) 1,560 mm × 1,910 mm × 640 mm (with arms unfolded and blades folded) 1,087 mm × 620 mm × 685 mm (with arms folded)	1,958 mm × 1,833 mm × 553 mm (with arms and blades unfolded) 1,232 mm × 1,112 mm × 553 mm (with arms unfolded and blades folded) 600 mm × 665 mm × 580 mm (with arms folded)
	Hovering Precision (with good GNSS signal)	With D-RTK enabled: ±10 cm (horizontal) and ±10 cm (vertical) With D-RTK disabled: ±0.6m (horizontal) and ±0.3m (vertical) (with the radar function enabled: ±0.1m)	With D-RTK enabled: ±10 cm (horizontal) and ±10 cm (vertical) With D-RTK disabled: ±0.6m (horizontal) and ±0.3m (vertical) (with the radar function enabled: ±0.1m)
	Hovering Endurance	16 min (@13,000 mAh & takeoff weight of 31 kg) 7.5 min (@13,000 mAh & takeoff weight of 51 kg)	19 min (@9500 mAh & takeoff weight of 16.8 kg) 8.7 min (@9500 mAh & takeoff weight of 26.8 kg)
Power System	Stator Size	100×33 mm	100×10 mm
	Maximum Power	4000 W/rotor	2000 W/rotor
	Propeller Diameter	54 inch	33 inch
	Rotor Quantity	4	4
Operation Tank	Operation Tank Volume	20L at full load	10L at full load
	Operation Load	20 kg at full load	10 kg at full load
Nozzles	Nozzle Model	LX8060SZ: 12 L/min	SX11001VS (standard): 1.8 L/min SX110015VS (optional): 2.4 L/min XR11002VS (optional): 3 L/min
	Nozzle Quantity:	2	4
	Atomized Particle Size	50 - 300 μm	SX11001VS: 130 - 250 μm SX110015VS: 170 - 265 μm XR11002VS: 190 - 300 μm
	Maximum Effective Spray Width	7 m (with a distance of 2.5 meters to crops)	3-5.5 m (with 4 nozzles and a distance of 1.5 to 3 meters to crops)
Water Pump	Model	Magnetic Drive Impeller Pump	Diaphragm Pump
	Maximum Flow	6 L/min x 2	1.5 L/min x 1

01 Product Introduction - Specifications

		T40/T20P	T30/T10
Omnidirectional Obstacle Avoidance Radar	Model	RD2484R	RD2424R
	Height Hold and Terrain Following	Height Measurement Range: 1 - 45 m Height Hold Range: 1.5 - 30 m Maximum Slope in Mountain Mode: 45 °	Height Measurement Range: 1 - 30 m Height Hold Range: 1.5 - 15 m Maximum Slope in Mountain Mode: 35 °
	Obstacle Avoidance System	Perceivable distance: 1.5 - 30 m FOV: 360° (horizontal), ±45° (vertical) Conditions of use: The relative flight height of the drone is above 1.5 m and the speed below 10 m/s Safe distance: 2.5 m (the distance between the tip of the blade and the obstacle after the drone is braked and hovering stably) Obstacle avoidance direction: Horizontally omnidirectional obstacle avoidance Perceivable distance: 1.5 - 30 m FOV: 45 ° Conditions of use: The relative distance between the aircraft and the obstacle above is greater than 1.5 m during takeoff, landing, and ascending. Safe distance: 2.5 m (the distance between the top of the aircraft and the obstacle after the drone is braked and hovering stably) Obstacle avoidance direction: Above the aircraft	Perceivable distance: 1.5 - 30 m FOV: 360° (horizontal), ±15° (vertical) Conditions of use: The relative flight height of the drone is above 1.5 m and the speed below 7 m/s. Safe distance: 2.5 m (the distance between the tip of the blade and the obstacle after the drone is braked and hovering stably) Obstacle avoidance direction: Horizontally omnidirectional obstacle avoidance
Backward Radar (T40/T20P) Upward Radar (T30/T10)	Model	RD2484B	RD2414U
	Obstacle Avoidance System	Perceivable distance: 1.5 -30 m FOV: Horizontal: ±60°, vertical: ±25° Conditions of use: The relative distance between the aircraft and the obstacle behind is greater than 1.5 m during takeoff, landing, and ascending. Safe distance: 2.5 m (the distance between the tip of the blade and the obstacle after the drone is braked and hovering stably) Obstacle avoidance direction: Behind the aircraft Perceivable distance: 1-45 m FOV: 45 ° Conditions of use: The relative distance between the aircraft and the obstacle below is 2-10 m during landing. Safe distance: 1.5 m (the distance between the bottom of the aircraft and the obstacle after the drone is braked and hovering stably) Obstacle avoidance direction: Below the aircraft	Perceivable distance: 1.5 - 10 m FOV: 80° Conditions of use: The relative distance between the aircraft and the obstacle above is greater than 1.5 m during takeoff, landing, and ascending. Safe distance: 2 m (the distance between the top of the aircraft and the obstacle after the drone is braked and hovering stably) Obstacle avoidance direction: Above the aircraft
Dual Vision Systems	Measurement Range:	0.4 - 25 m	
	Effective Sensing Speed:	≤10 m/s	
	FOV:	Horizontal: 90°, Vertical: 106 °	
	Operating Environment:	Surface with a clear pattern and adequate lighting	

01 Product Introduction — Material Recording (Repair)

Material Name	Picture	SN Position and Form	
T40		SN Position: Outside the aluminum tube of the middle frame Form: Sticker	
T20P		SN Position: Outside the aluminum tube of the middle frame Form: Sticker	

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02 Material Identification

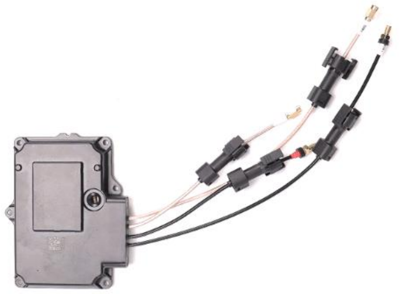
02 Material Identification — T40/T20P Universal Materials

FPV Module

Forward Vision Sensor Module

Omnidirectional Radar Module

Backward Radar Module



RF Module

Aerial-Electronics System Module

Spraying System Module

10033 Propulsion Motor

Auxiliary Light Module

ESC Module

RTK Antenna Module

Load Sensor Module

02 Material Identification — T40/T20P Universal Materials

Flowmeter Module

Impeller Pump

Centrifugal Spray Rod

Liquid Level Gauge

Liquid Level Gauge Sensor Module

Spraying Adapter Module

02 Material Identification — Specific Materials

Power Distribution Board Module

Cable Distribution Board Module

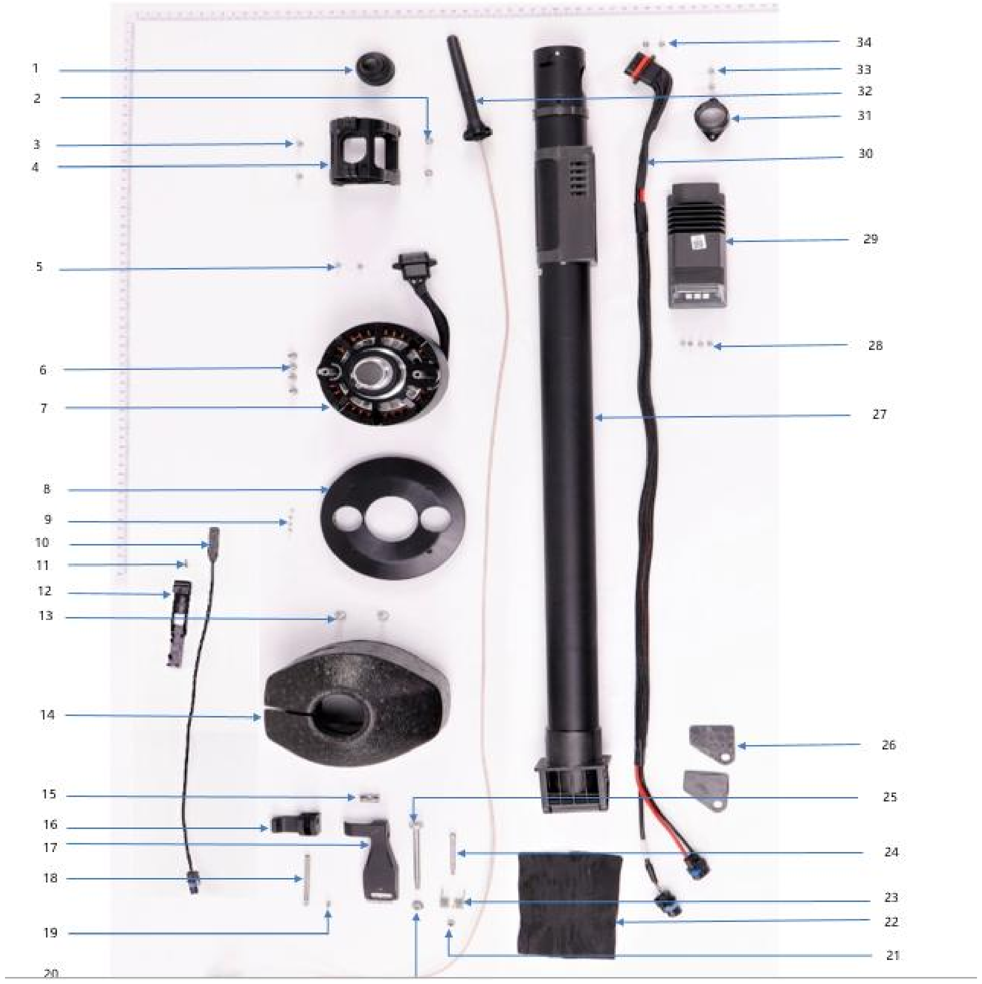
T40 Specific Materials

Power Distribution Board Module

Cable Distribution Board Module

T20P Specific Materials

02 Material Identification — BOM



No.	物料编码/Material Nur	Material Name	パ-
1	YC.JG.ZS002416	SDR Protective Sleeve	
2	YC.WJ.LL000232	M40-HC01000100-072028-5103-N_01	
3	YC.JG.LD000032	Motor Locating Pin	
4	YC.JG.QX002037	Motor Fixing Base	
5	YC.WJ.L00870	Screw M30-HC060060-55-85	
6	YC.ST.LL000148	M40-HC01400140-070039-4105-Y	
7	BC.AG.SS000566	10033 Propulsion Motor	
8	YC.JG.ZS002379	Motor Upper Cover	
9	YC.ST.LL000208	M20-TC00700060-033010-0004-Y	
10	YC.XC.XX000834	Aircraft Arm In-Position Detector Signal Cable	
11	YC.WJ.L01213	Screw T25-HC080080-45-21	
12	YC.JG.ZS001306	On-Site Measurement Bracket	
13	YC.WJ.LL000213	M30-HT01930060-110025-5103-Y	
14	YC.JG.FP000019	Motor Cover	
15	YC.JG.QX001243	Locking Piece_Handle Connecting Rod Shaft	
16	YC.JG.MY000353	Locking Piece_Handle Rubber Sleeve	
17	BC.AG.SS000605	Locking Piece Handle	
18	YC.JG.QX001242	Locking Piece_Connecting Rod	
19	YC.WJ.L00871	Screw M30-HC100100-55-85	
20	YC.ST.LM000030	Screw M50-HN080070-0003	
21	YC.WJ.L00961	M4 Stainless Locknut	
22	FL.SC.BB000001	Acetate Fabric	
23	YC.JG.TT000095	Locking Piece Spring	
24	YC.JG.LD000023	Left and Right Locking Piece Screw Bolt	
25	YC.JG.LD000026	Aircraft Arm Fixing Screw Bolt	
26	YC.JG.ZS002324	Aircraft Arm Friction Plate	
27	BC.AG.SS000571	M1 Aircraft Arm (Front Right)	
28	YC.WJ.L00870	Screw M30-HC060060-55-85	
29	BC.AG.SS000563	ESC Module	
30	YC.XC.XS000014	M1&M2 Aircraft Arm ESC Composite Cable	
31	YC.JG.ZS002291	Spray Lance Fixing Base	
32	YC.DZ.AA000275	SDR Antenna	
33	YC.WJ.L00871	Screw M30-HC100100-55-85	
34	YC.ST.LL000122	Screw M25-HC00780040-040020-5108-N	

Note: For more details, refer to the corresponding Product Material Information Compilation.

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03 Repair Strategy

03 Repair Strategy

Product	Module	Repair Strategy for Dealers
Aircraft	Aerial-Electronics System Module	Replace the whole module.
	Barometer	Replace the whole module.
	Spraying System Module	Replace the whole module.
	RF Module	Replace the whole module.
	Cable Distribution Board Module	Replace the whole module.
	Auxiliary Light Module	Replace the whole module.
	Load Sensor Module	Replace the whole module.
	Forward Vision Sensor Module	Replace the whole module.
	FPV Module	Replace the whole module.
	Aircraft Arm Module	Replace the propulsion motor.
		Replace the ESC.
		Repair the remaining components
	Omnidirectional Radar Module	Replace the whole module.
	Backward Radar	Replace the whole module.
	Spray Tank-Flowmeter Module	Replace the whole module.
	Spray Tank-Spraying Adapter Module	Replace the whole module.
	Spray Tank-Impeller Pump	Repair the pump.
	Spray Tank-Liquid Level Gauge	Replace the whole module.
	Spray Tank-Liquid Level Gauge Sensor	Replace the whole module.
	Spray Tank-Main Body	Repair the main body.
	Power Distribution Module	Replace the whole module.
	RTK Antenna Module	Replace the whole module.
	Landing Gears	Repair the landing gears.
	Centrifugal Spray Rod	Repair the spray rod.
	Airframe	Repair the airframe.
	CCW and CW Propellers	If one of the two propellers on one motor is damaged, replace the two propellers. The corresponding propeller gasket needs to be replaced during the repair.

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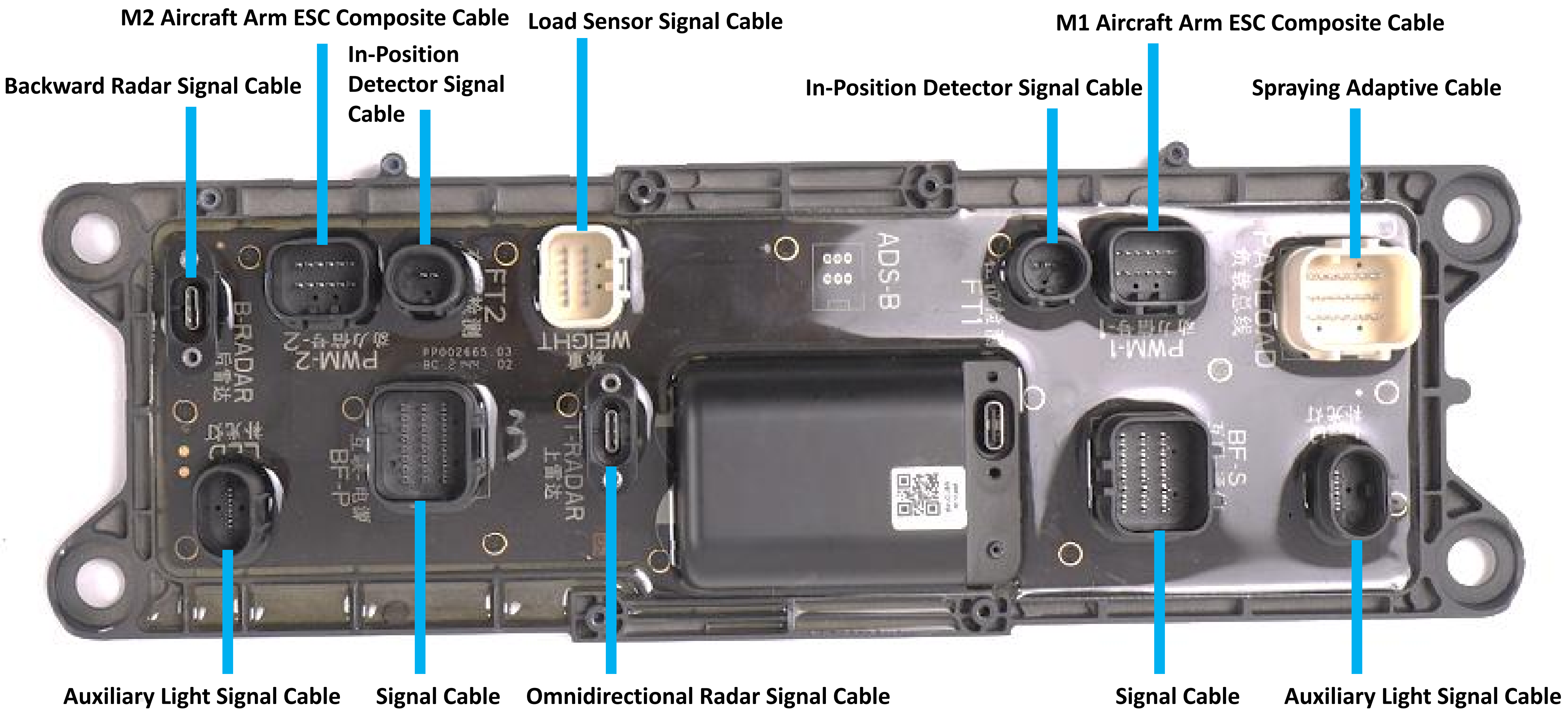
04 Disassembly and Assembly Notes

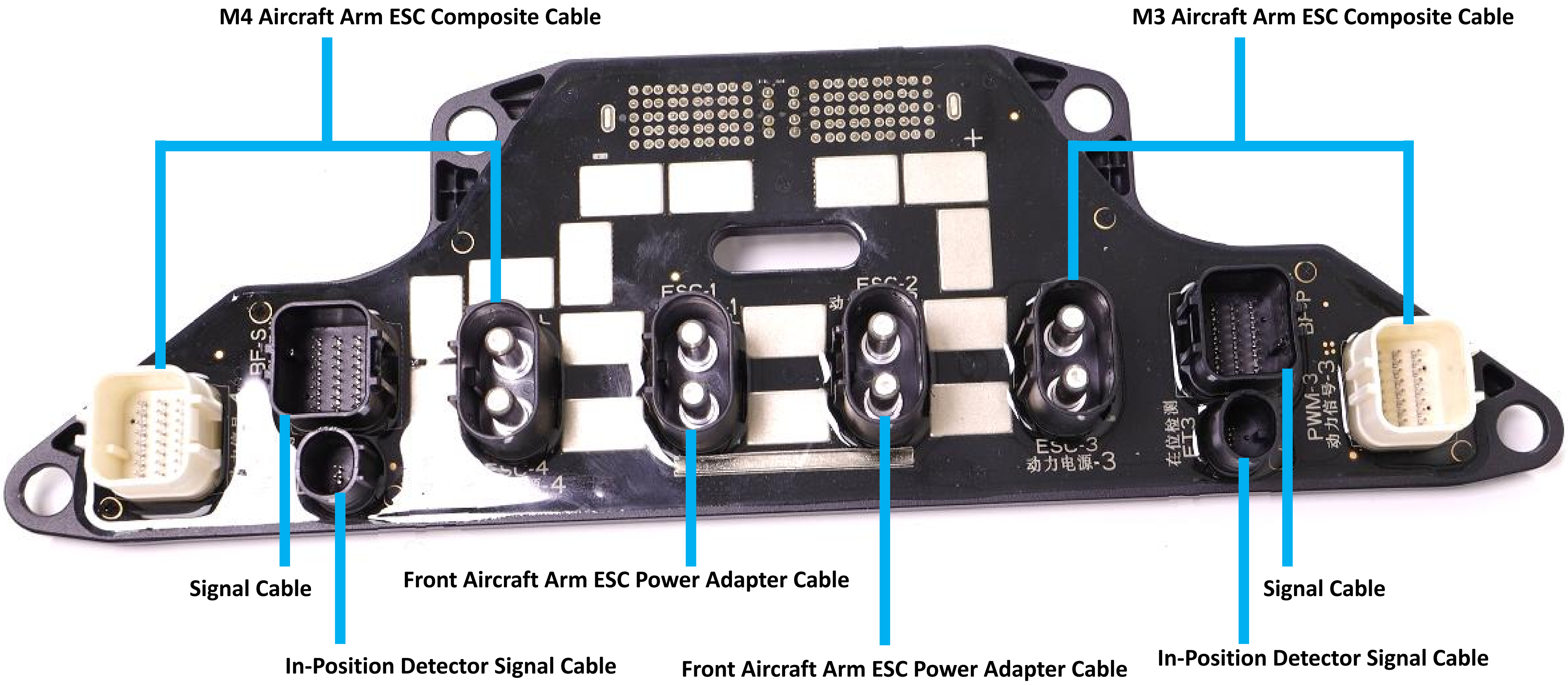
04 Disassembly and Assembly Notes

◆ Cable Connectors on the T40 Cable Distribution Board:



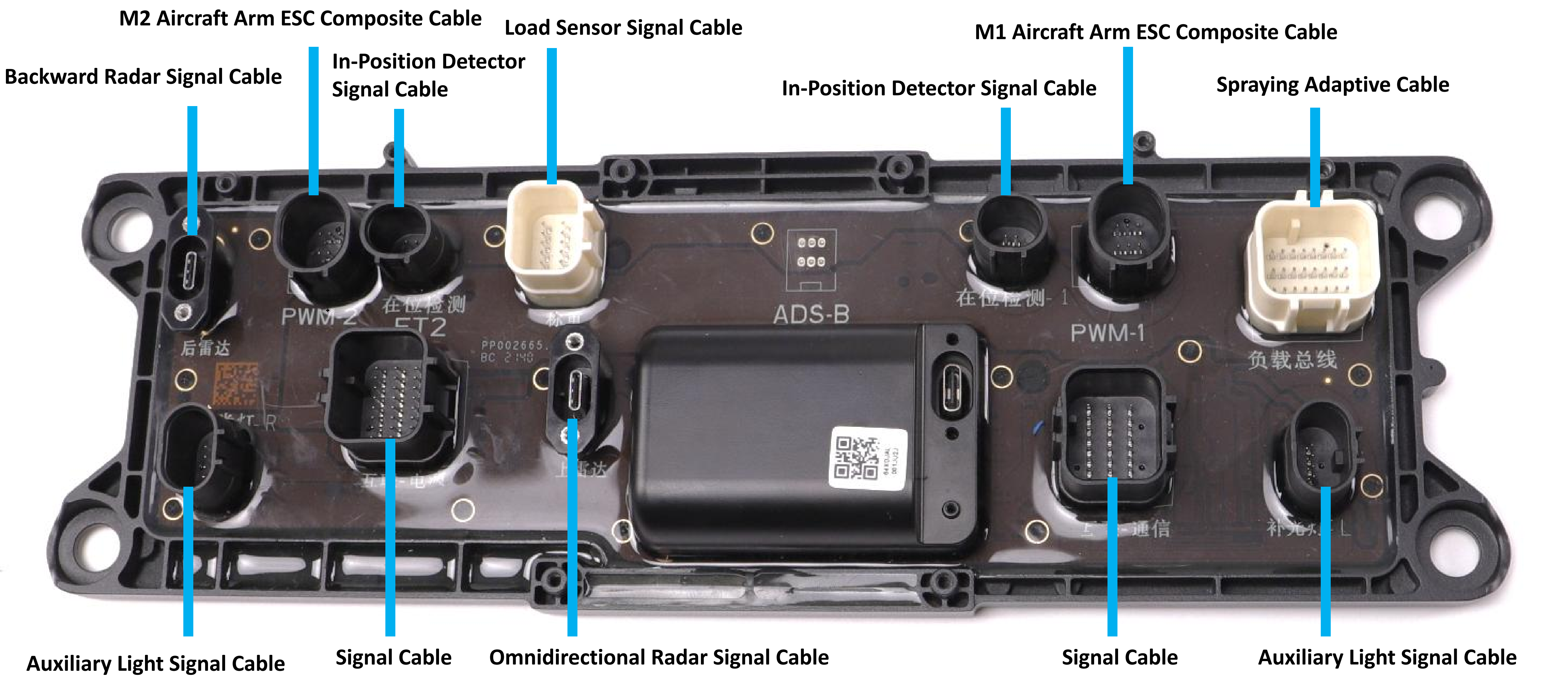
04 Disassembly and Assembly Notes

◆ Cable Connectors on the T40 Power Distribution Board:



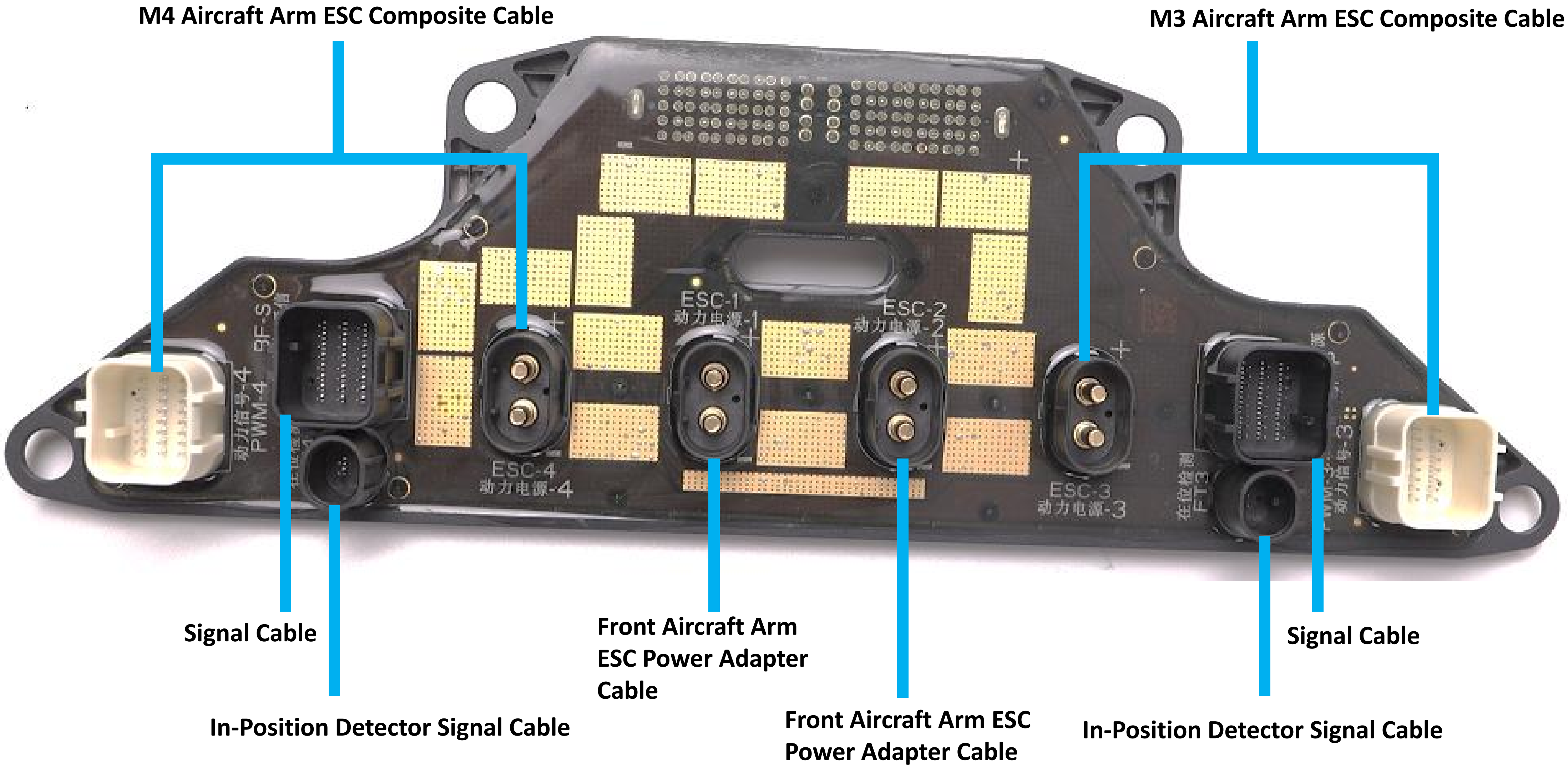
04 Disassembly and Assembly Notes

◆ Cable Connectors on the T20P Cable Distribution Board:



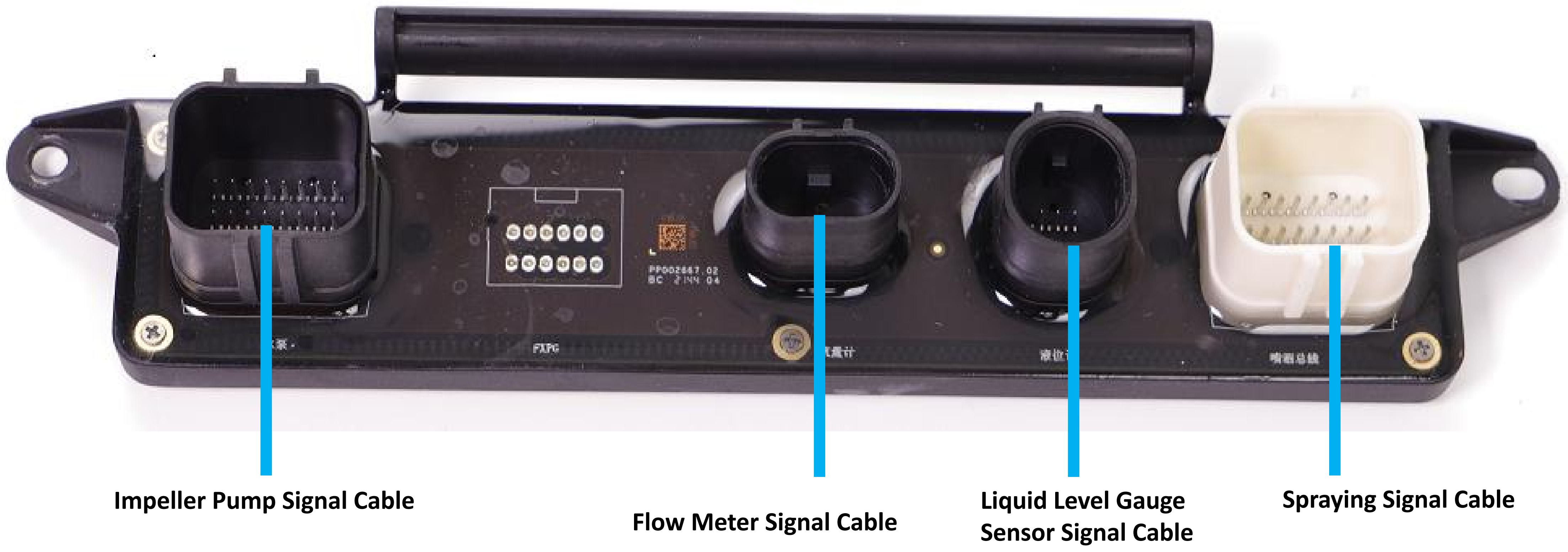
04 Disassembly and Assembly Notes

◆ Cable Connectors on the T20P Power Distribution Board:

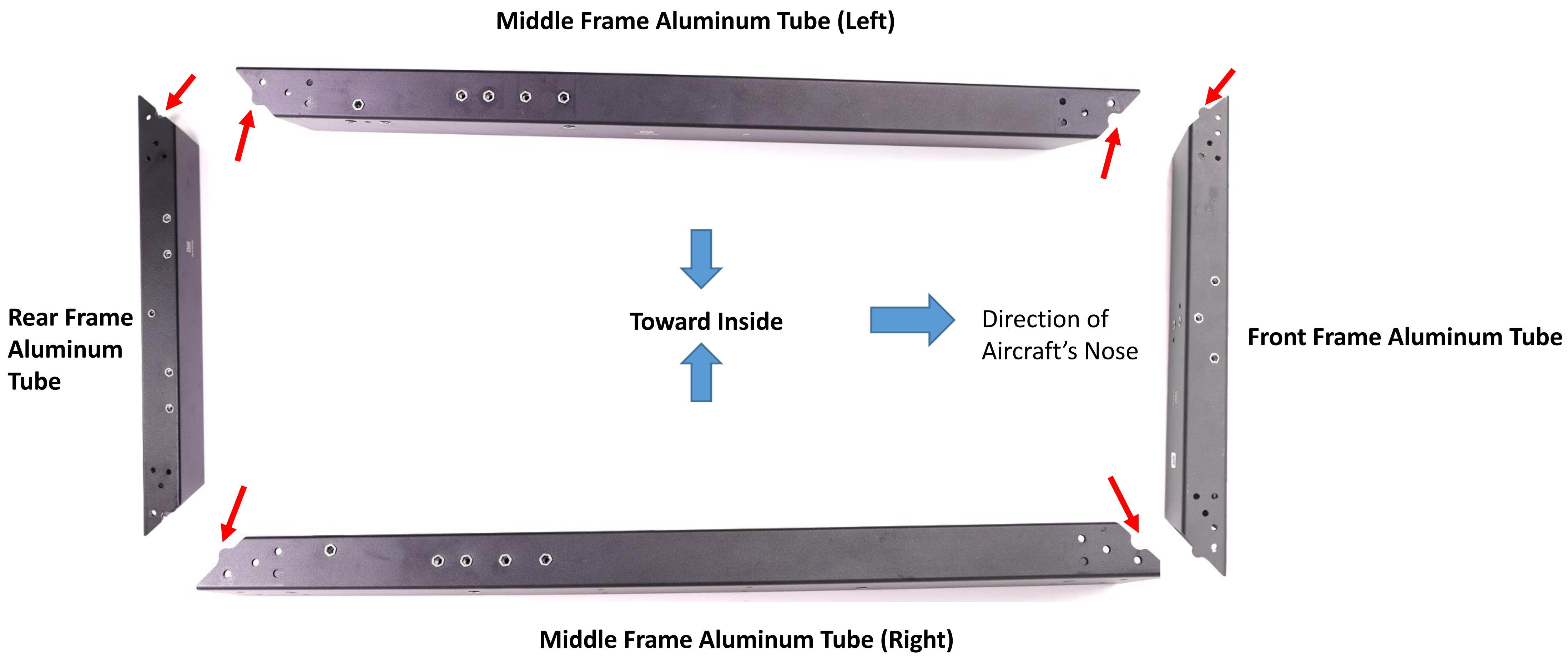


04 Disassembly and Assembly Notes

◆ Cable Connectors on the Spraying Adapter Module



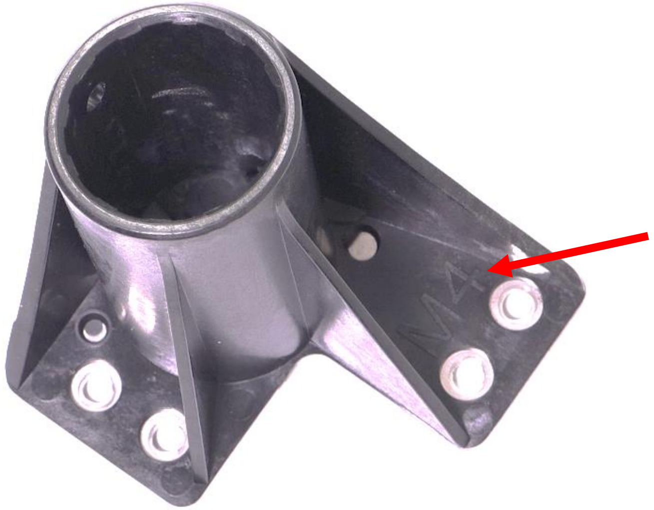
04 Disassembly and Assembly Notes — Aircraft



T40 and T20P drones use the same material of the aluminum tube. Please note that the surface of the aluminum tube with slotted part and raised part is the front surface, and the left and right aluminum tubes with the slotted side face the direction of the aircraft's nose, and the front and rear aluminum tubes with the raised part are mounted to the front frame.

04 Disassembly and Assembly Notes — Aircraft

After the aircraft arm lock is replaced, fasten the lever. The tension value needs to be tested (acceptable tension range: 90±10N).



Landing Gear Fixing Piece

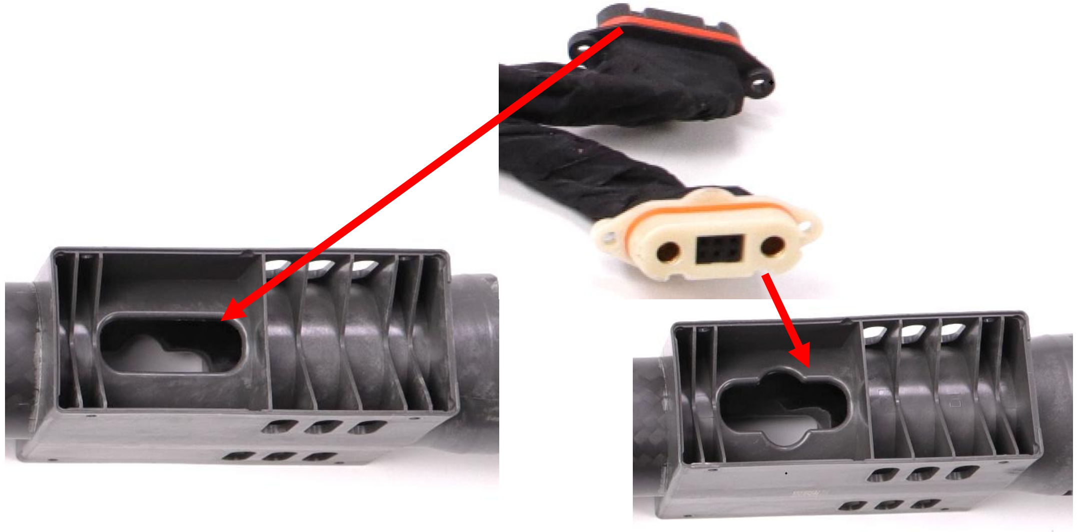
T40 and T20P use the same materials of the landing gear fixing piece with the mark of M1-M4 that are mounted below the corresponding M1-M4 aircraft arms.

04 Disassembly and Assembly Notes — Aircraft

T40 M1 Aircraft Arm Connector

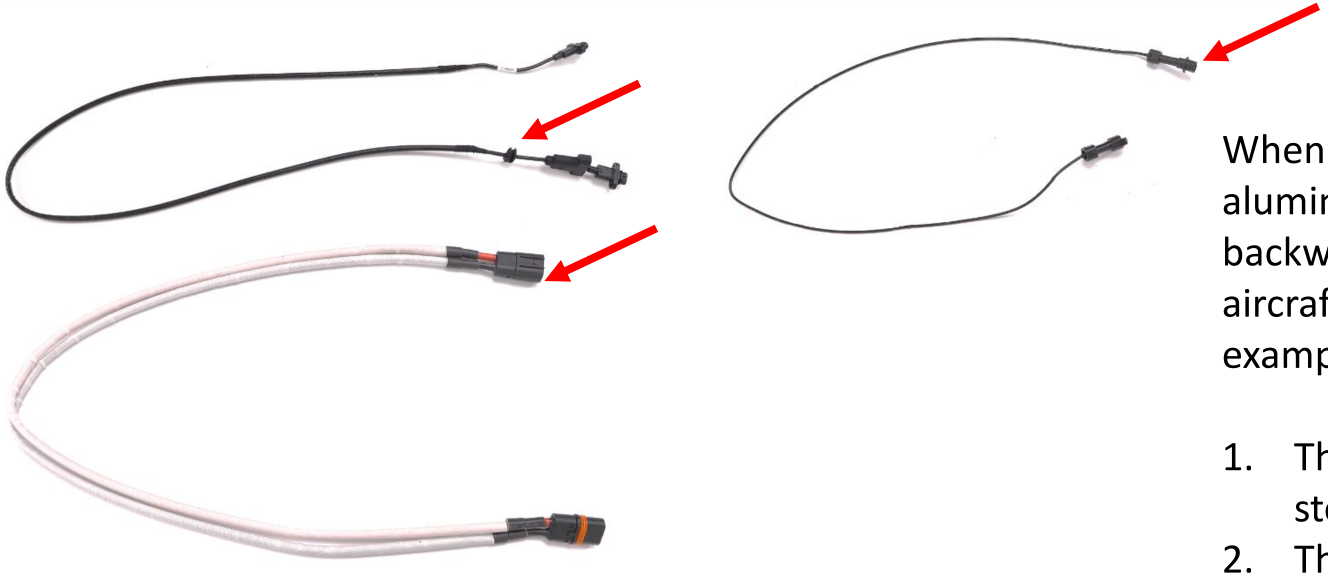
T20P M1 Aircraft Arm Connector

1. There is M1 CCW or M5 CW mark on the T40 aircraft arm connector.
2. There is M1 CCW mark on the T20P aircraft arm connector.



The ESC bracket of the T40 aircraft arm has different shapes of slots on both sides. Make sure that the side with the "petal" shape slot needs to face downwards, and the aircraft ESC composite cable connector should be passed through the slot.

04 Disassembly and Assembly Notes — Aircraft



When passing the cables through the left and right aluminum tubes, pay attention to the direction of the backward radar signal cable/RTK coaxial cable/the front aircraft arm ESC adapter cable (take T40 cables as an example as shown on the left).

1. The backward radar signal cable with the rubber stopper should face the direction of the rear frame.
2. The RTK coaxial cable/front aircraft arm ESC adapter cable "female " connector needs to face the direction of the front frame.

The spray rod adapter shaft and spray rod fixing nut are not easy to be removed after they are mounted.
1. If the spray rod fixing nut needs to be replaced separately, destructively remove it.
2. If the spray rod fixing nut needs to be replaced separately, replace the spray rod fixing nut together.

04 Disassembly and Assembly Notes — Aircraft

To use the T40 spray rod hose with the T20P drone, **the non-threaded end part** (close to the centrifugal motor) of the hose should be cut off about 8 cm. Otherwise, the spray rod hose cannot be connected to the aircraft arm hose.

04 Compatibility List

Flight Battery	T40	T20P	T30
T40 Intelligent Flight Battery	Supported	Supported; Increase the flight time.	Supported; Increase the flight time.
T20P Intelligent Flight Battery	Not Supported	Supported	Not Supported
T30 Intelligent Flight Battery	Supported; needs to reduce the payload weight.	Supported; Increase the flight time.	Supported
Battery Station	**T40 Intelligent Flight Battery**	**T20P Intelligent Flight Battery**	**T30 Intelligent Flight Battery**
D12000iE Multifunctional Inverter Generator	Supported	Supported (power adaptation)	Supported (power adaptation)
T40 Battery Station	Supported	Supported (power adaptation)	Supported (power adaptation)
D6000i Multifunctional Inverter Generator	Supported; Slow charging	Supported	Supported; Slow charging
T20P Battery Station	Supported; Slow charging	Supported	Supported; Slow charging
DJI D9000i Multifunctional Inverter Generator	Supported; Slow charging (firmware update required)	Supported (firmware update required)	Supported
T30 Intelligent Battery Station	Supported; Slow charging (firmware update required)	Supported (firmware update required)	Supported
Remote Controller	**T40**	**T20P**	**T30**
T40 DJI RC Plus	Supported	Supported	Not Supported
T20P DJI RC Plus	Supported	Supported	Not Supported
Propellers	**T40**	**T20P**	**T30**
T40 Drone Propellers (Upper)	Supported	Supported	Not Supported
T40 Drone Propellers (Lower)	Supported	Not Supported	Not Supported
T20P Drone Propellers	Supported (Upper Propellers)	Supported	Not Supported

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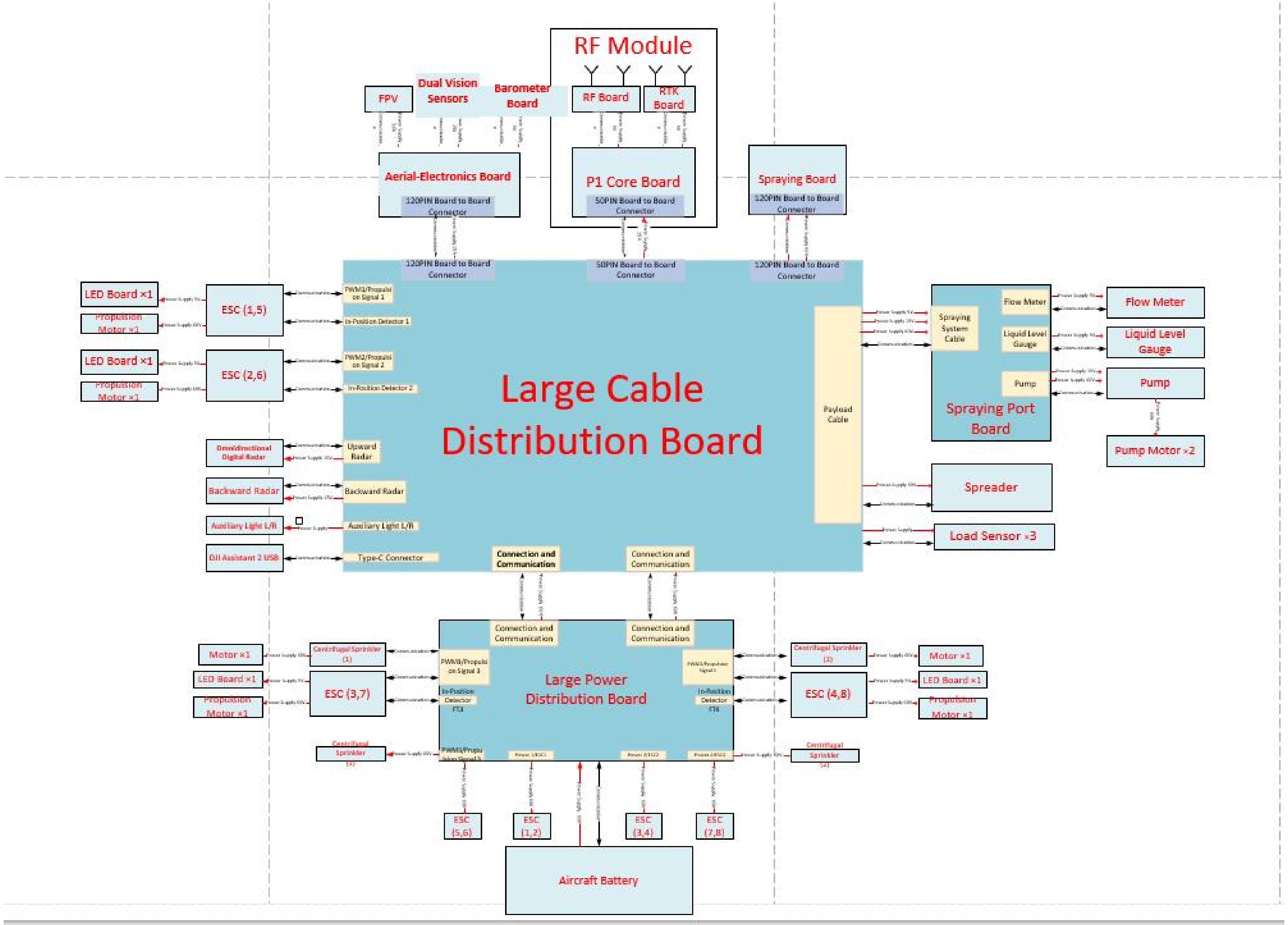
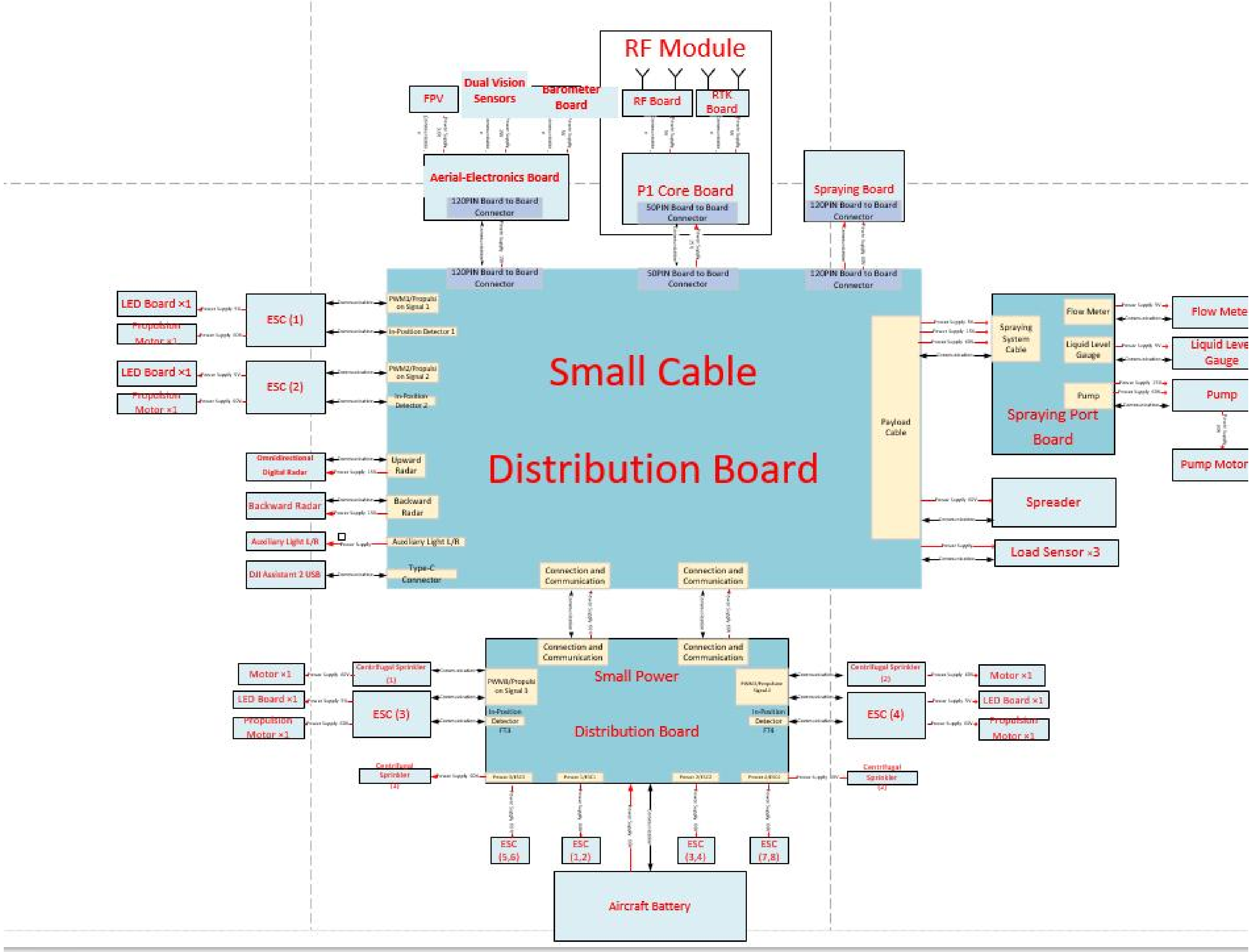
05 Principles Diagram and Boards Introduction

05 Principles Diagram — T40



05 Principles Diagram — T20P



05 Boards Introduction

System	Module/Board	Function
Aerial-Electronics System	Aerial-Electronics Module	The aerial-electronics module is the core module controlling the flight controller. It integrates the flight algorithm and flight controlling functions into one module, which contains the compass and barometer.
	Barometer	Receives the current barometric pressure signal, sends the barometric pressure information to the aerial-electronics board, and detects the aircraft's current altitude information.
Propulsion System	Motor	The motor works with propellers to blow wind downwards and make the aircraft fly upwards. It also helps with spraying.
	ESC Module	The ESC module converts the throttle information to the motor's rotation speed information, controls the motor rotation, sends the motor information, and triggers the overcurrent protection.
	LED	LED indicates the aircraft's current working condition of the aircraft. Users can know the aircraft's current working condition by the LED's color.
	Power Distribution Board	Distributes the power to the four/eight motors, which is the transfer station for the battery and power supply.
	Cable Distribution Board	Connects the cables and modules.
Spraying System	Spraying System Module	The control system of the aircraft's core spraying function. It controls the pump, centrifugal nozzle, and spraying volume.
	Liquid Level Gauge	Works with the load sensor to measure the current liquid amount in the tank to notify the customer whether to add liquid.
	Load Sensor	Works with the liquid level gauge to measure the current liquid amount in the tank to notify the customer whether to add liquid.
	2-Channel Flow Meter	It is essential for high-precision spraying that detects the pump's spraying speed. Therefore, the pump's spraying accuracy can be controlled.
	Delivery Pump	The impeller pump is the exerciser of the aircraft spraying system, which transmits pesticides to the centrifugal nozzle by rotating and sprays pesticides out.
	Centrifugal Sprinkler	The liquid is atomized and sprayed from the spinner through the centrifugal force.
Positioning System	RF Module (Baseband Board, RF Board, RTK/GPS Board)	As a flight controller system to provide the aircraft position, direction, and speed of the aircraft to ensure the flight stability, and the efficiency and accuracy of the operation. It can track GPS L1/L2, GLONASS F1/F2, BDS B1/B2, and Galileo E1/E5b frequency bands at the same time. It has positioning and orientation functions. The GPS module uses the antenna of RTK module to receive GPS L1, and Glonass F1 frequency band signals.
	Active Antenna Board	Amplify and adjust the weak GPS signal so that the RTK/GPS board can get better satellite signals.
Obstacle Avoidance System	Omnidirectional Digital Radar	A sensor for obstacle avoidance and navigation of aircraft. Responsible for investigating the external environment to prevent the aircraft from hurting people and crash.
	Backward Radar	Detects the environment behind the aircraft to prevent collisions with obstacles.
	Dual Vision Sensors	Detects obstacles through vision sensors and works with the radars to make the drone avoid obstacles, and provides the terrain following function.
Image Transmission system	FPV & Gimbal	Converts optical signals into digital signals, and transmits them to customers through image transmission. Aerial mapping is available that the fields captured by FPV are transmitted to the remote controller for reconstruction.
	Auxiliary Light	The auxiliary lights are installed on both sides of the front of the aircraft's nose, which is responsible for lighting in the dark environment.
	RF Module (Baseband Board, RF Board, RTK/GPS Board)	Contains baseband board, RF board; RF module is used to process the image transmission signal, and connect the aircraft and remote controller.
Battery System	Battery	Batteries are the source of energy for flight and spraying. It is an energy storage device.

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06 Test Plan and Fixture

06 Test Plan and Fixture — Aircraft

No.	Module	Test Item	Content	Device/Fixture	Test Environment	Software	Damage Assessment Required	Repair Test Required
1	Aircraft	Short-Circuit Test	Connect the positive and negative poles to confirm whether there is a risk of short circuit.	Multimeter and Regulated Power Supply	/	/	Yes	Yes
2		Firmware Update	1. Check whether all modules are mounted in place. 2. Update the firmware to the latest version.	/	Not related	DA2-MG	No	Yes
3		Dual Vision Sensors Calibration	Calibrate the vision sensors (if there is a prompt stating that the vision sensor is malfunctioning).	Vision Calibration Board YC.SJ.G00509.01 YC.BZ.J01205.01	DJI Assistant 2 software	DA2-MG	No	Yes
4		App Functionality Test	1. RTK satellite searching test 2. Vision sensor obstacle detection test 3. Auxiliary light check 4. Load sensor data download 5. FPV camera data download 6. Aircraft arm in-position check 7. Water pump calibration	Not related	Not related	App	Yes	Yes
5 6		Flight Test	1. Linking test 2. Radar obstacle avoidance check 3. Manual flight	Not related	Not related	App	Yes	Yes
7	Impeller Pump	Spray Test	1. Check if the flow can reach 6 L/min. 2. Check whether the impeller pump is water leakage.	Not related	Not related	App	Yes	Yes
	Centrifugal Spray Rod	Spray Test	1. Check whether the spray button is functioning. 2. Check the spraying speed and check whether the dial can be used to control the spraying normally. 3. Check whether the spray rod is water leakage.	Not related	Not related	App	Yes	Yes

1. After the gimbal and aerial-electronics system module are replaced, it is required to download the camera data.
2. Vision sensor calibration is required after the dual vision sensors module is replaced.
3. Download the load sensor data after the load sensor and aerial-electronics system module are replaced.

06 Test Plan and Fixture — Test Plan Requird After Component Replacement — Aircraft

Test	Module	Test Item	Items Need to be Tested after Part Replacement																				
			Auxiliary Light	FPV	Load Sensor	Cable Distribution Board	Power Distribution Board	Omnidirectional Obstacle Avoidance Radar	Backward Radar	Aerial-Electronics System	Spraying System	Barometer	Forward Vision Sensor	RTK Antenna	RF	ESC	Motor	Impeller Pump	Centrifugal Spray Rod	Flow Meter	Spraying Adapter Board	SDR Antenna	Liquid Level Gauge Sensor
Aircraft Test	Aircraft	Short-Circuit Test	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
		Firmware Update	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
		Dual Vision Sensors Calibration				√				√			√										
		App Functionality Test	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
		Flight Test	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√

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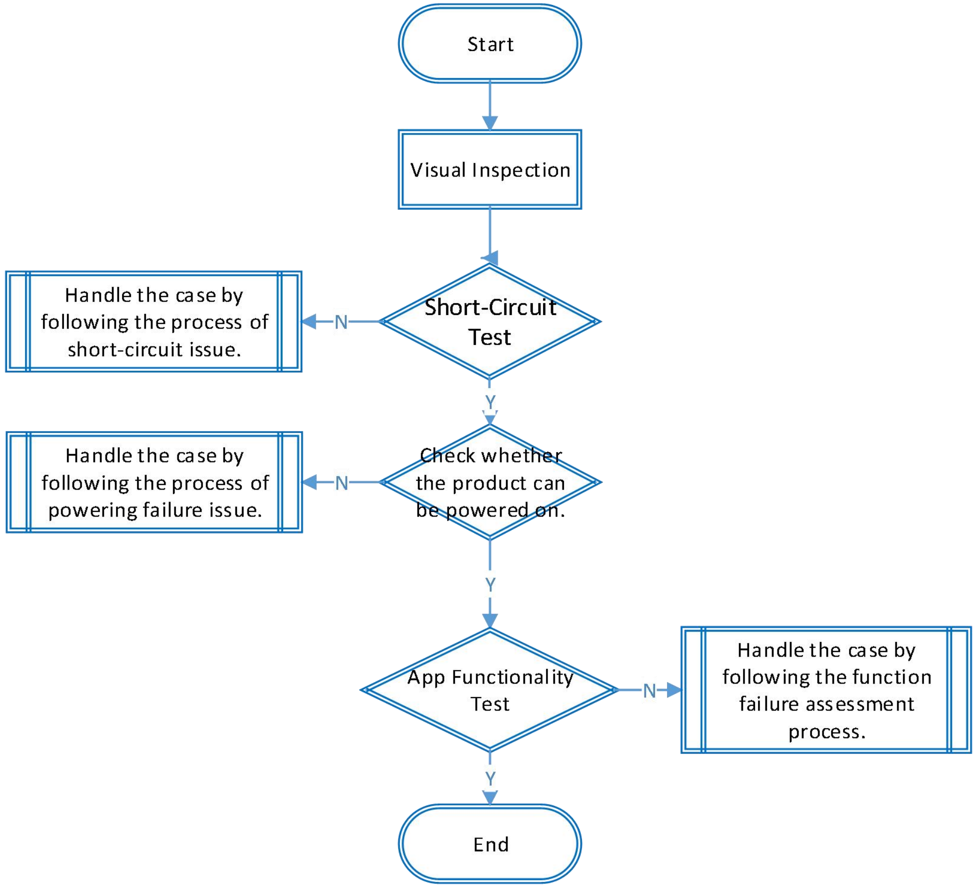
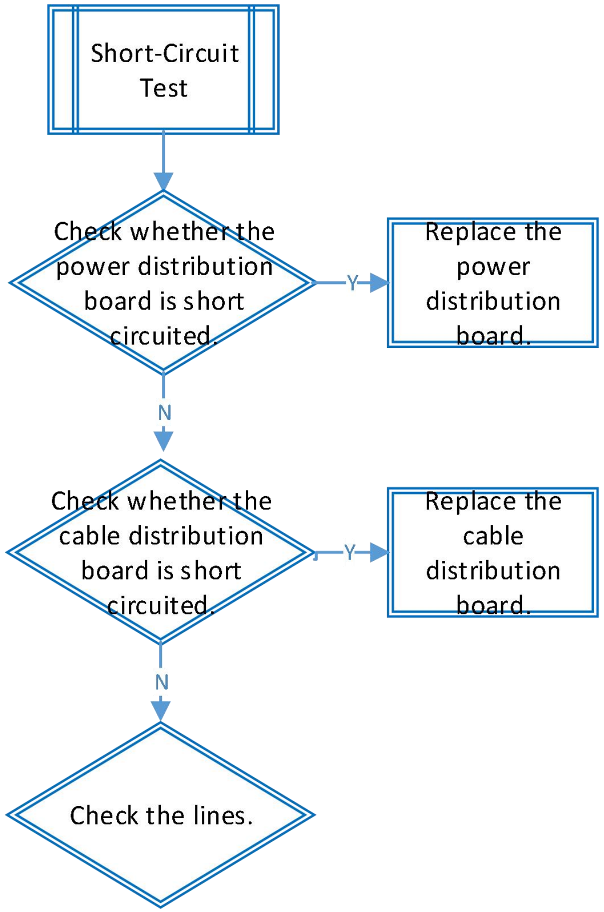
07 Malfunction Assessment and Analysis

07 Malfunction Assessment and Analysis — Common Issues

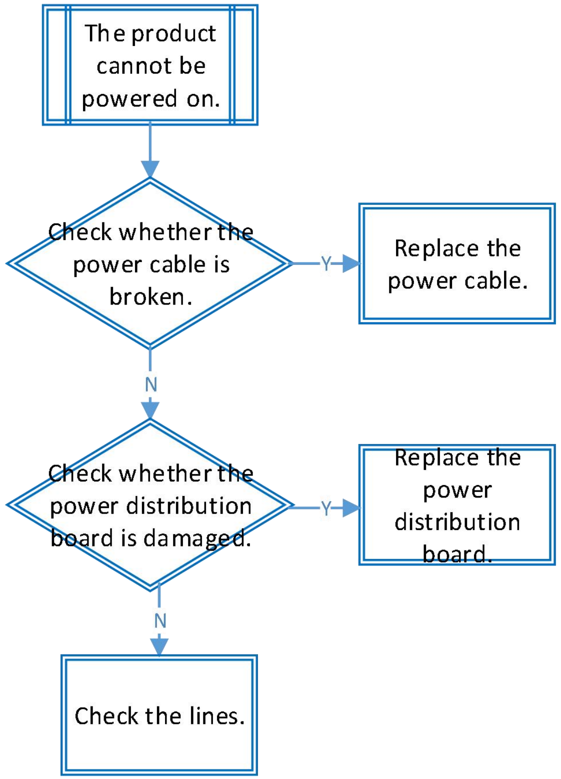
Using the app for damage assessment.



07 Malfunction Assessment and Analysis — Common Issues



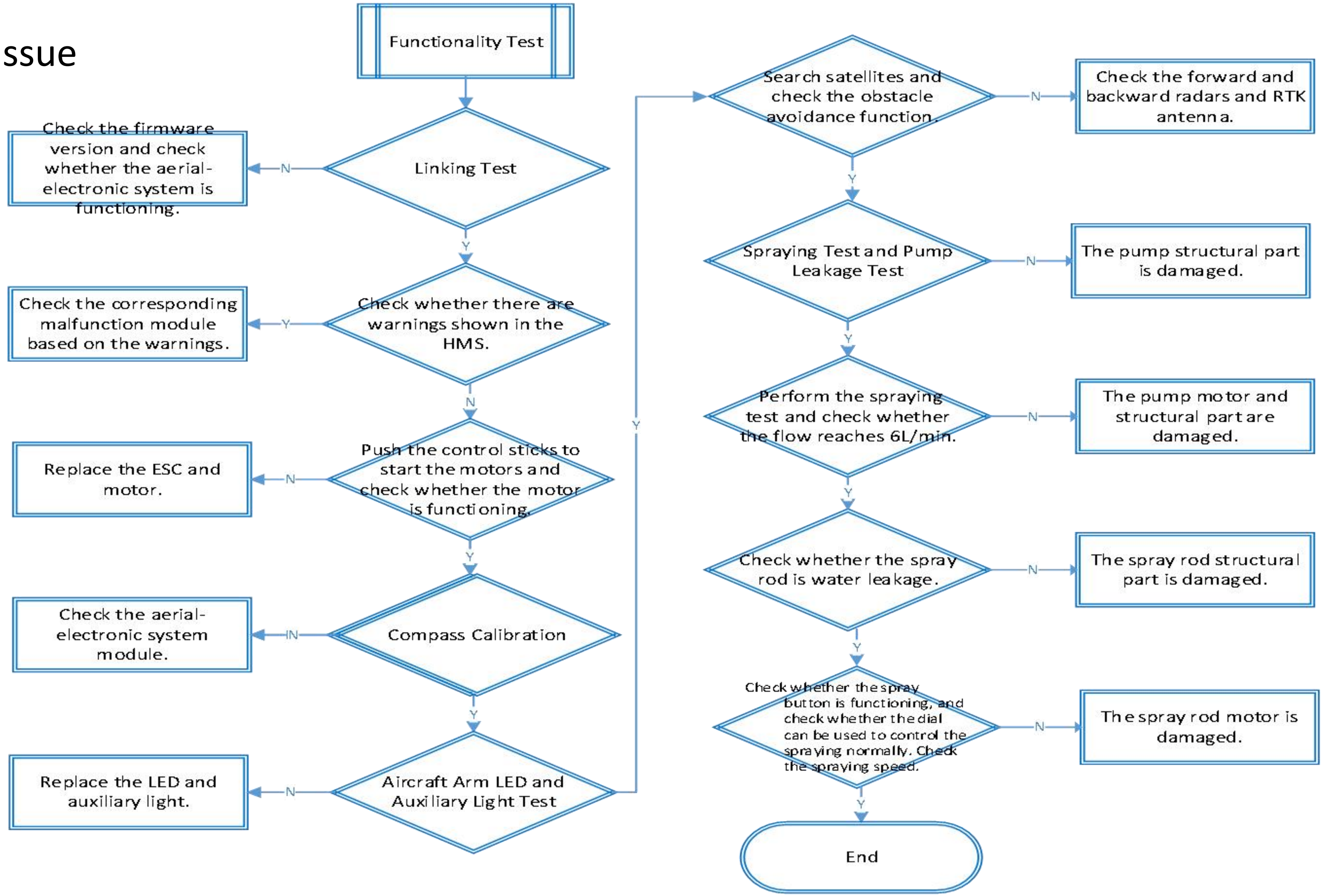
Short-Circuit Issue



Power on Failure Issue

07 Malfunction Assessment and Analysis — Common Issues

Function Failure Issue



07 Malfunction Assessment and Analysis — Water Damage

Check if the liquid damage indicator has turned red, and if any boards are moldy.

<table>
<tr><th colspan="2">Liquid Damage Indicator</th><th rowspan="2">Board Status</th><th rowspan="2">Assessed a Water Damaged or not?</th></tr>
<tr><th>Liquid Damage Indicator</th><th>Liquid Damage Indicator</th></tr>
<tr><td rowspan="4">With liquid damage indicator</td><td rowspan="2">Turned red</td><td>Corrosion/Water Stain/Water Mark</td><td>Yes</td></tr>
<tr><td>No Corrosion/Water Stain/Water Mark</td><td>No</td></tr>
<tr><td rowspan="2">Color unchanged/Not turned White After Soaked in Liquid</td><td>Corrosion/Water Stain/Water Mark</td><td>Yes</td></tr>
<tr><td>No Corrosion/Water Stain/Water Mark</td><td>No</td></tr>
<tr><td rowspan="2">Without liquid damage indicator</td><td rowspan="2">NA</td><td>Corrosion/Water Stain/Water Mark</td><td>Yes</td></tr>
<tr><td>No Corrosion/Water Stain/Water Mark</td><td>No</td></tr>
</table>

Thank You